

GEOG 531

Quantitative Methods in Geographic Research

Fall 2026

T Th 2:50 – 4:05 PM

Callcott Social Science Center 302

Instructor: Varun Goel

E-mail: varungoel@sc.edu

Office: Callcott Social Science Center 111

Student Support Hours: **T Th 1:30 – 2:30 pm** & by appointment

Academic Bulletin Description

Quantitative Methods in Geographic Research

Course Description

What is the distribution of a different phenomenon (such as disease cases, air pollution, political beliefs etc.) across space (and time)? Why does a particular spatial pattern exist, and what underlying spatial or ecological processes may be driving those patterns? How do we design a study or analysis that incorporates spatial thinking? Answering these questions needs to make use of quantitative methods (or statistical analysis) with geographic data. This course is for advanced undergraduates or graduate students in Geography and related disciplines who want to gain necessary spatial statistical analysis knowledge and skills for geographic research.

This course will deal with the nature of geographical data sets, and statistical measures and models commonly used by geographers to describe spatial variations and patterns, distributions, and relationships among geographical data. Each student will be given opportunities to apply these techniques to geographical datasets and questions (from a wide variety of domains spanning environmental, social, and health sciences) through hands-on lab exercises, homework assignments, and a student-driven final project. Additionally, students will also gain general insights on how to execute a sound quantitative research idea, including basics of study design, sampling, and scientific writing and communication, and gain experience with a popular coding language used in quantitative research.

Prerequisites

There are no prerequisites for this course. The course IS NOT about how to solve equations or learning how to code. However, any previous experience with GIS, or R (or another programming language) will be helpful. Knowledge of basic algebra will also be helpful for concepts in the second half of the course. As such, the course is designed to get you “up to speed” on these concepts through additional resources, tutorials, and labs.

Learning Outcomes

By the end of this course, students will be able to:

1. **Explain** core concepts in spatial thinking, spatial data structures, and the statistical foundations of geographic analysis.
2. **Apply** descriptive and inferential statistical techniques to explore spatial variations, distributions, and relationships in geographic data.

3. **Interpret** spatial patterns and **evaluate** possible underlying processes using spatial statistical models.
4. **Use** a statistical programming language (e.g., R) and GIS software to go through multiple steps of a quantitative geographic research project from data cleaning to statistical modeling.
5. **Communicate** spatial statistical results effectively through visualizations and written explanations.
6. **Integrate** spatial analysis into interdisciplinary research questions from environmental, social, and health sciences, among others.
7. Critically read and evaluate quantitative methods and statistical claims in published geographic research, including the interpretation of statistical significance, model coefficients, assumptions, uncertainty, and causal claims.
8. **Produce** a final research product that synthesizes course concepts into an original, data-driven geographic analysis.

Course Materials

There are no required textbooks for this course, and we will be using several openly available sources and reading materials that will be placed on blackboard or on the course website.

However, I do recommend a couple of books that are helpful for modern quantitative thinking and geographic research:

- 1) [Introduction to Statistical Problem Solving in Geography \(4th Edition\)](#)
- 2) [Introduction to Modern Statistics \(2nd Edition\)](#)

All course materials will comply with copyright/fair use policies.

Technology Requirements

Lecture presentations, links to articles, assignments, and other course materials will be available on the [course website](https://geog531.netlify.app) [https://geog531.netlify.app]. To participate in learning activities and complete assignments, you will need access to a working computer, reliable Internet access and a USC email account, a current Internet browser, a word processing program, and reliable data storage for your work. Students will heavily use the R programming language and RStudio for this course, primarily through a cloud-based version called [Posit](#). Detailed instructions will be provided, and all software will also be available on computers in the class lab.

Course Policies and Procedures

Attendance Policy

Regular attendance is strongly encouraged because this is an active, discussion- and lab-based course, and many class meetings will include hands-on exercises, reading checks, methods critiques, guest speakers, or collaborative activities. Attendance itself is not graded. Instead, students earn credit for work completed during class as described under In-Class Exercises, Labs, Reading Checks & Methods Discussions. Your four lowest in-class activity scores will automatically be dropped, providing flexibility for occasional illness, personal obligations, travel, or other circumstances.

- Because this flexibility is built into the course, individual in-class activities generally cannot be made up. Students remain responsible for material covered during an absence. Extended circumstances, university-protected absences, and approved accommodations will be handled individually in accordance with university policy.
- Regardless of whether you have an excused or unexcused absence, you will be expected to complete a makeup assignment in either case. Makeup assignments and their instructions will be provided by the instructor.

In Class Smartphone and Computer Use Policy

Unless instructed, your electronic devices, including your laptop should be placed on silent and should be put away in your pocket or bag. Lecture slides will be available on the course website before the class for you to print and take notes with a pen if you would like. If there is a serious need to leave your phone on vibrate (such as a family emergency), please let me know. If you happen to leave the class to accept a phone call, then I will have advance knowledge of the potential emergency.

Late Work Policy

All assignments will be posted on the course website with detailed instructions including the due dates. All assignments must be submitted electronically prior to the due date/time. Instructions for electronic submission will be provided. Unless you have an [excused absence](#) or have a previous agreement with me, late submissions will be penalized 10% for each day overdue.

Academic Integrity

From *Academic Responsibility* – The Honor Code (STAF 6.25): “It is the responsibility of every student at the University of South Carolina to adhere steadfastly to truthfulness and to avoid dishonesty in connection with any academic program. A student who violates, or assists another in violating the Honor Code, will be subject to university sanctions.” Students who engage in this type of behavior will receive, at a minimum, a “0” (zero) for the work in question and risk failure for the course. Other sanctions may also be pursued.

Plagiarism

Using the words or ideas of another as if they were one’s own is a serious form of academic dishonesty. If another person’s complete sentence, syntax, key words, or the specific or unique ideas and information are used, one must give that person credit through proper citation.

Generative Artificial Intelligence (AI) Policy

Generative AI tools such as ChatGPT, Gemini, Claude etc. should not be used in the completion of course assignments unless the instructor specifically authorizes their use. You will be informed as to when and how these tools may be used, along with guidance for attribution if/as needed. Any use of generative AI tools outside of these parameters constitutes plagiarism and will be treated as such.

Class Conduct

Professionalism will be expected at all times. Because the university classroom is a place designed for the free exchange of ideas, we must show respect for one another in all circumstances. We will show respect for one another by exhibiting patience and courtesy in our exchanges. Appropriate language and restraint from verbal attacks upon those whose perspectives differ from your own is a minimum requirement. Courtesy and kindness are the norm for those who participate in my class.

Course Assignments and Grading

General Assignment Information

- All coursework (e.g., assignments, projects) and rubrics will be available on blackboard or course website.
- All assignments will be due, and all exams administered on the day indicated on the course schedule.
- All assignments will be posted and submitted on Posit or blackboard.
- All in-class labs will be administered in class.

Evaluation

Final grades in this course will be based on in-class exercises and engagement, a midterm exam, homework assignments, and a final Quantitative Geography Showcase.

1) In-Class Exercises, Labs, Reading Checks & Methods Discussions (250 points)

Most class meetings will include a short, graded activity designed to help you practice quantitative reasoning and engage actively with course material. Activities may include R exercises, short labs, open-note reading checks, critiques of statistical methods or claims in published papers, small-group problem solving, guest-speaker responses, and project workshops. These activities emphasize meaningful engagement and application rather than high-stakes performance. Your four lowest in-class activity scores will automatically be dropped when calculating this category. Because these drops provide built-in flexibility for occasional absences or difficult days, individual in-class activities generally cannot be made up.

Reading checks and methods critiques are part of this same in-class category. Assigned methodological readings may address topics such as statistical significance, causal interpretation, model assumptions, reproducibility, and common errors in interpreting regression results. Reading checks will generally be brief and may be open-note; their purpose is to prepare students to discuss and apply the arguments rather than memorize details.

2) Midterm Exam (150 points)

The midterm exam will assess students' understanding of core concepts and their ability to interpret and reason about quantitative geographic analyses covered in the first half of the course. This exam will be administered as a pen and paper in-class assignment.

3) Homework Assignments (450 points, 150 each)

Students will be asked to complete 3 structured homework assignments throughout different points of the semester. The purpose of these assignments is to help you apply techniques and concepts learned in class to major contemporary real world data sets. These are individual assignments and will be assessed based on completion and a grading rubric that will be distributed by the instructor.

4) Final Quantitative Geography Showcase (150 points)

All students will complete a final Quantitative Geography Showcase that synthesizes course concepts through an original, data-driven geographic analysis. Students may pursue an independent geographic research question or, when offered, work from a common data challenge. The core deliverables will include a reproducible analysis (for example, in Quarto), a polished communication product appropriate to a real audience, a brief presentation of the research story, and a short reflection on what the analysis can and

cannot support. Project ideas will be developed through low-stakes checkpoints and in-class project workshops during the semester. Students will present their final project during the scheduled final exam day and time at the end of the semester.

Graduate Students: Graduate students will complete the same core Showcase but will be held to higher expectations for methodological justification and critical evaluation. In addition to the core deliverables, graduate work should engage relevant research literature, justify analytical choices relative to plausible alternatives, demonstrate appropriate attention to assumptions and robustness or sensitivity, and discuss how methodological decisions affect inference. Graduate students may also be asked to briefly defend their methodological choices during the Showcase discussion.

Grading Scale

All graded assignments will be returned to you on Blackboard. You are strongly encouraged to check your grades regularly. A final letter grade will be assigned based on percentages.

Assignment Weights	Points
In-Class Exercises, Labs, Reading Checks & Methods Discussions	250
Midterm	150
Homework Assignments	450
Final Quantitative Geography Showcase	150
Total	1000

Grading Scale

900 - 1000 = A
 850 - 899 = B+
 800 - 849 = B
 750 - 799 = C+
 700 - 749 = C
 650 - 699 = D+
 600 - 649 = D
 0 - 599 = F

Course Schedule

The instructor reserves the right to make changes to this schedule; students will be informed in advance where practical.

Day	Date	Topic	Activities
T	18-Aug	- Course Introduction - Intro to quantitative thinking for geographic questions	Introduction to R, Rstudio and Posit Cloud
Th	20-Aug		

Day	Date	Topic	Activities
T	25-Aug	- Observations and Variables - Intro to Geographic Data	- Analyzing and summarizing data structures in R - Importing/Exporting Spatial Data
Th	27-Aug		
T	1-Sep	- Study designs - Population vs samples - Sampling methods - Geographic considerations in study designs	- Sampling and Randomization in R - Reproducibility and Replicability
Th	3-Sep		
T	8-Sep	- Descriptive statistics - Basics of Data visualization	HW 1: Descriptive Problem solving in Geography
Th	10-Sep		
T	15-Sep	- Descriptive Spatial Statistics - Exploratory (spatial) Data analysis	Visualizing Multivariate relationships
Th	17-Sep		
T	22-Sep	- Point pattern analysis - Area pattern analysis	Exploratory Spatial Data Analysis workflow in R
Th	24-Sep		
T	29-Sep	- Foundations of inference - Probability, sampling distributions, bootstrapping - Hypothesis testing and confidence intervals	HW2: Exploratory Spatial Data Analysis
Th	1-Oct		
T	6-Oct	- Causation vs Correlation - Statistical Inference techniques	- Regression modeling and diagnostics - Directed Acyclic Graphs
Th	8-Oct		
T	13-Oct	Midterm Exam	
Th 15-Oct Fall Break (No Classes)			
T	20-Oct	- Ordinary Least squares regression - Multiple regression	Regression modeling and diagnostics
Th	22-Oct		

Da y	Date	Topic	Activities
T	27-Oct	- Generalized Linear Regression - Hierarchical Regression Model	Interpreting and visualizing regression results
Th	29-Oct		
T 3-NOV General Election Day (No Classes)			
Th	5-Nov	Spatial Regression	HW3: Inferential Statistics
T	10-Nov	Geographical weighted regression	Communicating quantitative results and findings
Th	12-Nov		
T	17-Nov	- Dimensionality reduction in Spatial Data - Unsupervised Clustering Analysis - Supervised Clustering Analysis	Final Showcase practice
Th	19-Nov		
TTH 24-26 Nov Thanksgiving Break (No Classes)			
T	01-Dec	- Spatial Prediction and Forecasting	Course wrap-up
Th	03-Dec		
M	8-Dec	Final Quantitative Geography Showcase 4:00 – 7:00 pm	

Academic Success

Your instructor is committed to helping you achieve the best in this course. Below is a list of valuable support resources available at USC. Please reach out to me for assistance in accessing any of these support services.

Disability Services

The [Student Disability Resource Center](http://www.sa.sc.edu/sds/) (SDRC) (<http://www.sa.sc.edu/sds/>) empowers students to manage challenges and limitations imposed by disabilities. In order to receive reasonable accommodations from me, you must be registered with the Student Disability Resource Center (1705 College Street, Close-Hipp Suite 102, Columbia, SC 29208, 803-777-6142). Any student with a documented disability should contact the SDRC to make arrangements for appropriate accommodations. Once registered, students with disabilities are encouraged to contact me (within the first week of the semester) to discuss the logistics of any accommodations needed to fulfill course requirements.

Student Success Center

In partnership with USC faculty, the [Student Success Center](https://sc.edu/about/offices_and_divisions/student_success_center/index.php) (SSC) (https://sc.edu/about/offices_and_divisions/student_success_center/index.php) offers a number of programs to help you better understand your course material and to support your path to success. SSC programs are facilitated by professional staff, graduate students, and trained undergraduate peer leaders who have previously excelled in their courses. Resources available to you in this course may include:

- **Peer Tutoring:** You can make a one-on-one appointment with a [Peer Tutor](http://www.sc.edu/success) (www.sc.edu/success). Drop-in Tutoring and Online Tutoring may also be available for this course. Visit their website for a full schedule of times, locations, and courses.
- **Supplemental Instruction (SI):** SI Leaders are assigned to specific sections of courses and hold three weekly study sessions. Sessions focus on the most difficult content being covered in class. The SI Session schedule is posted through the SSC website each week and will also be communicated in class by the SI Leader.
- **Peer Writing:** Improve your college-level writing skills by bringing writing assignments from any of your classes to a Peer Writing Tutor. Similar to Tutoring, you can visit the website to make an appointment, and to view the full schedule of available drop-in hours and locations.
- **Success Consultations:** In Success Consultations, SSC staff assist you in developing study skills, setting goals, and connecting to a variety of campus resources. Throughout the semester, I may communicate with the Student Success Center regarding your progress, which indicates your instructor is concerned about your progress in this course. If contacted by the Student Success Center, please schedule a Success Consultation right away. Referrals are not punitive, and any information shared by your professor is confidential and subject to FERPA privacy laws. Student Success Center services are offered to all USC undergraduates at no additional cost. Please call 803-777-1000, visit www.sc.edu/success, or come to the Student Success Center in the Thomas Cooper Library (Mezzanine Level) to check schedules and make appointments.

Writing Center

This course has many writing assignments. The University [Writing Center](http://artsandsciences.sc.edu/write/university-writing-center) (<http://artsandsciences.sc.edu/write/university-writing-center>) is an important resource you should use! It's open to help any USC student needing assistance with a writing project at any stage of development. The main Writing Center is in Byrnes 703.

University Library Resources

[University Libraries Resources](http://sc.edu/libraries) (sc.edu/libraries)

- University Libraries has access to books, articles, subject specific resources, citation help, and more. If you are not sure where to start, assistance is available at [Ask a Librarian!](https://sc.edu/libraries/ask) (<https://sc.edu/libraries/ask>).
- Remember that if you use anything that is not your own writing or media (quotes from books, articles, interviews, websites, movies – everything) you must cite the source in MLA (or other appropriate and approved) format.

Counseling Services

The University offers [Counseling and crisis services](https://sc.edu/about/offices_and_divisions/student_health_services/medical-services/counseling-and-psychiatry/index.php)

(https://sc.edu/about/offices_and_divisions/student_health_services/medical-services/counseling-and-psychiatry/index.php) as well as outreach services, self-help, and frequently asked questions.

Mental Health

If stress is impacting you or getting in the way of your ability to do your schoolwork, maintain relationships, eat, sleep, or enjoy yourself, then please reach out to any of USC's mental health resources. Most of these services are offered at no cost as they are covered by the Student Health Services tuition fee. For all available mental health resources, check out [Student Health Services Mental Health](https://www.sc.edu/about/offices_and_divisions/health_services/mental-health/index.php)

(https://www.sc.edu/about/offices_and_divisions/health_services/mental-health/index.php) and the quick reference list below.

- Wellness Coaching can help you improve in areas related to emotional and physical well-being (e.g., sleep, resiliency, balanced eating and more) – schedule an appointment at (803) 777-6518 or on [MyHealthSpace](https://myhealthspace.ushs.sc.edu/login_dualauthentication.aspx) (https://myhealthspace.ushs.sc.edu/login_dualauthentication.aspx)
- Access virtual self-help modules via [Therapy Assistance Online \(TAO\)](https://us.taoconnect.org/register) (<https://us.taoconnect.org/register>) – see [TAO registration instructions](https://www.sc.edu/about/offices_and_divisions/health_services/medical-services/counseling-and-psychiatry/online-support/index.php) (https://www.sc.edu/about/offices_and_divisions/health_services/medical-services/counseling-and-psychiatry/online-support/index.php).
- Access additional articles and videos on health and wellness topics on the Wellness Hub, thriveatcarolina.com, or by downloading the [CampusWell](https://www.campuswell.com/) (<https://www.campuswell.com/>) app and searching for University of South Carolina.
- Counseling & Psychiatry offers individual and group counseling and psychiatric services – schedule an appointment at (803) 777-5223 or on [MyHealthSpace](https://myhealthspace.ushs.sc.edu/login_dualauthentication.aspx) (https://myhealthspace.ushs.sc.edu/login_dualauthentication.aspx).
- Access the 24-hr Mental Health Support Line at (833) 664-2854.
- Access an anonymous [mental health screening program](https://www.uscscscreening.org/welcome.cfm?access=website) (<https://www.uscscscreening.org/welcome.cfm?access=website>).

Interpersonal Violence

Interpersonal violence - including sexual harassment, relationship violence, sexual assault, and stalking - is prohibited at USC. Faculty, staff, and administrators encourage anyone experiencing interpersonal violence to speak with someone, so they can get the necessary support and USC can respond appropriately. If you or someone you know has been or is currently impacted by interpersonal violence, you can find the appropriate resources at the [Sexual Assault and Violence Intervention & Prevention](https://sc.edu/safety/interpersonal-violence/index.php) (SAVIP) (<https://sc.edu/safety/interpersonal-violence/index.php>) website.

As faculty, I must report all incidents of interpersonal violence and sexual misconduct, and thus cannot guarantee confidentiality. Please know that you can seek [confidential resources](https://sc.edu/safety/interpersonal-violence/index.php) (<https://sc.edu/safety/interpersonal-violence/index.php>). If you want to make a formal report, you can [report here](https://cm.maxient.com/reportingform.php?UnivofSouthCarolina&layout_id=25) (https://cm.maxient.com/reportingform.php?UnivofSouthCarolina&layout_id=25) or contact the institution's Title IX Coordinator, or one of the Deputy Title IX Coordinators listed on the SAVIP website. You can also file a police report by contacting USC Police at 803-777-4215.

Diversity and Inclusion

The university is committed to a campus environment that is inclusive, safe, and respectful for all persons, and one that fully embraces the Carolinian Creed. To that end, all course activities will be conducted in an atmosphere of friendly participation and interaction among colleagues, recognizing and appreciating the unique experiences, background, and point of view each student brings. You are expected at all times to apply the highest academic standards to this course and to treat others with dignity and respect.

Diversity, Ethics, and the Carolinian Creed

This course works to foster a climate free of harassment and discrimination, and it values the contributions of all forms of **diversity**. The decision to enter university and pursue advanced study is a choice that entails commitment to personal **ethics** expressed in the [Carolinian Creed](https://sc.edu/about/offices_and_divisions/student_affairs/our_initiatives/involvement_and_leadership/carolinian_creed/index.php) (https://sc.edu/about/offices_and_divisions/student_affairs/our_initiatives/involvement_and_leadership/carolinian_creed/index.php): "I will discourage bigotry, while striving to learn from differences in people, ideas and opinions." Likewise, the [Student Code of Conduct](http://www.sc.edu/policies/ppm/staf626.pdf) (STAF 6.26: <http://www.sc.edu/policies/ppm/staf626.pdf>) stresses, "The University of South Carolina strives to maintain

an educational community that fosters the development of students who are ethical, civil and responsible persons.”

Title IX and Gendered Pronouns

This course affirms equality and respect for all gendered identities and expressions. Please don't hesitate to correct me regarding your preferred gender pronoun and/or name if different from what is indicated on the official class roster. Likewise, I am committed to nurturing an environment free from discrimination and harassment. Consistent with Title IX policy, please be aware that I as a responsible employee am obligated to report information that you provide to me about a situation involving discrimination, sexual harassment, or assault.